HUMPHREY VIRTUAL FARMS LLC // LEVEL-5 SOVEREIGN C2 ENCLAVE
CAGE: 1AHA8 | UEI: S1M4ENLHTDH5 | STATUTORY: OK Title 61 / HB 2992 | DOC ID: HVF-TRADEWINDS-10-26-3976
CLASSIFICATION: SOVEREIGN LEVEL-5 EXECUTIVE DELIVERABLE // PUBLIC/PRIVATE PARITY

TRADEWINDS RESUBMISSION DOSSIER // DOCKET 10-26-3976 (RESUBMISSION OF 9-26-3703)

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DOD CDAO TRADEWINDS SOLUTIONS MARKETPLACE // RESUBMISSION DOSSIER & VIDEO SCRIPT
ACTIVE DOCKET: 10-26-3976 (RESUBMISSION OF 9-26-3703) | READINESS: TRL 4 TESTBED
CONTRACTOR: Humphrey Virtual Farms LLC | CAGE: 1AHA8 | UEI: S1M4ENLHTDH5

PART 1: POINT-BY-POINT RESOLUTION OF PRIOR DOCKET 9-26-3703 ASSESSOR FEEDBACK

[AF-01] CRITIQUE : Clarify Technology Readiness Level (TRL) claims vs. physical field hardware deployment.
RESOLUTION: Explicitly baselined at TRL 4 (Laboratory Software Emulation Testbed & Mathematical Proof-of-Concept) running on air-gapped local loopback (127.0.0.1, 0% external cloud dependency). Physical PCB fabrication and TRL 6/7 field range trials are funded milestones under our proposed 12-Month, $1.485M Phase-1 Prototype OTA.

[AF-02] CRITIQUE : Substantiate sub-microsecond (36.75ns) Kinetic Guillotine and zero-trust cybersecurity claims with empirical proof.
RESOLUTION: Distinguished the 36.75ns physical MOSFET cutoff (28.5W / 24V DC) as our SPICE/mathematical hardware design target for Phase-1 PCB fabrication, while demonstrating live on-screen execution of 84/84 passing automated software test suites and 17/17 blocked adversarial red-team penetration vectors in our local loopback software testbed.

[AF-03] CRITIQUE : Clarify NIST SP 800-171 / CMMC / SPRS compliance posture and Government Data Rights.
RESOLUTION: Completed 110-control architectural mapping under NIST SP 800-171 Rev 2 and NIST SP 800-82 Rev 2 across our Three-Brain software testbed; scheduled formal third-party C3PAO / DoD SPRS audit as a funded Phase-1 OTA milestone; and locked 100% Small Business Background IP retention with Government Purpose Rights (GPR) under DFARS 252.227-7018.

[AF-04] CRITIQUE : Demonstrate clear DoD Problem Alignment, Defense Customer Transition Path, and Commercial Viability (Section VII.e).
RESOLUTION: Demonstrated live Three-Brain C2 & SCADA isolation for DoD tactical microgrids, base utilities, and civil works (USACE ERDC / DIU / CDAO), backed by a 4-milestone $1.485M Phase-1 Prototype OTA schedule ($150K M1A Mobilization) under 10 U.S.C. Section 4022 and an auditable 1,768-module Public GitHub blueprint.

PART 2: SECTION VII.e COMPLIANT VIDEO TELEPROMPTER SCRIPT (5 SEGMENTS)

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>>> SEGMENT 1 (0:00 - 0:45) // CORPORATE IDENTITY & PRIOR ASSESSOR FEEDBACK RESOLUTION

RUBRIC: Mandatory Resubmission Compliance Gate (Addressing Docket 9-26-3703 Assessor Feedback)

SCRIPT: I am Jeffery Humphrey, Founder, President, and CEO of Humphrey Virtual Farms LLC, CAGE Code 1AHA8, UEI S1M4ENLHTDH5. This video is our formal resubmission for Project Ebony under Tradewinds Docket 10-26-3976, directly addressing every item of Assessor Feedback from our prior submission 9-26-3703. First, we have explicitly baselined Project Ebony at TRL 4--a laboratory software emulation testbed running 100% offline on local loopback 127.0.0.1 with zero external cloud dependency. Second, we clarify that our 36.75-nanosecond Kinetic Guillotine cutoff is our mathematical and SPICE hardware design target for Phase-1 PCB fabrication, while our software testbed is proven live on screen today across 84 of 84 passing automated unit tests and 17 of 17 blocked red-team cyber-intrusion vectors. Third, our 110-control NIST 800-171 and 800-82 architectural mapping is complete, with formal C3PAO and SPRS certification funded inside our proposed 12-month, $1.485 million Phase-1 Prototype OTA.

>>> SEGMENT 2 (0:45 - 1:30) // DOD PROBLEM STATEMENT & OPERATIONAL IMPACT (SECTION VII.e.1)

RUBRIC: Problem Identification, Operational Relevance & Warfighter Impact

SCRIPT: The Department of Defense faces a critical operational vulnerability: tactical microgrids, forward operating base utilities, autonomous UAS ground stations, and civil infrastructure SCADA controllers rely on cloud-tethered software and probabilistic AI that fail under electronic warfare, bandwidth denial, or PLC bus injection. When an adversary compromises a traditional SCADA bus or spoofs telemetry, cloud latency cannot prevent kinetic equipment destruction.

>>> SEGMENT 3 (1:30 - 3:00) // TECHNICAL SOLUTION & LIVE TRL 4 TESTBED DEMO (SECTION VII.e.2)

RUBRIC: Technical Merits, Innovation, Architecture & Live Demonstration

SCRIPT: Project Ebony solves this through a deterministic Three-Brain architecture. Brain One enforces deterministic physical boundary invariants and models our 36.75-nanosecond Kinetic Guillotine cutoff at 28.5 Watts and 24 Volts DC. Brain Two runs air-gapped structural, pavement, and telemetry diagnostics locally on 127.0.0.1 with zero cloud calls and zero authority to bypass Brain One safety limits. Brain Three seals every sensor frame and operator directive into a bare-metal SQLite Write-Once-Read-Many SHA-256 Merkle DAG vault across 52 verified database tables. On screen right now, you see our live TRL 4 software testbed executing all 84 automated unit test suites with zero failures, blocking all 17 adversarial red-team penetration vectors including automatic single-letter tamper zeroization, and verifying 100% cryptographic parity across our 1,959 Private and 1,768 Public Python modules.

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>>> SEGMENT 4 (3:00 - 4:00) // DIFFERENTIATION, DATA RIGHTS & CYBER POSTURE (SECTION VII.e.3)
RUBRIC: Competitive Differentiation, IP Sovereignty & Cybersecurity
SCRIPT: Unlike commercial cloud wrappers or black-box AI vendors, Humphrey Virtual Farms LLC is a 100% sovereign small business prime. All 1,959 Python modules, Three-Brain interlocks, and our 1,420-question Sovereign Field Academy workforce bank were engineered 100% at private expense. Under DFARS 252.227-7018, we retain background IP while granting the Department of Defense full Government Purpose Rights for funded Phase-1 prototype deliverables.

>>> SEGMENT 5 (4:00 - 4:45) // ACQUISITION PATHWAY & PHASE-1 OTA MILESTONES (SECTION VII.e.4)
RUBRIC: Commercialization, Transition Strategy & Payable Milestone Schedule
SCRIPT: Project Ebony is ready for immediate procurement under a 12-month, $1.485 million Phase-1 Prototype Other Transaction Agreement pursuant to 10 U.S.C. Section 4022, beginning with a $150,000 Day 15-to-30 mobilization tranche to procure ruggedized edge hardware and oscilloscope test benches, advancing from TRL 4 software emulation to TRL 6 and TRL 7 field demonstration by Month 12. Thank you.